

Ex. No.: 7

IPC USING SHARED MEMORY

Aim:

To write a C program to do Inter Process Communication (IPC) using shared memory between sender process and receiver process.

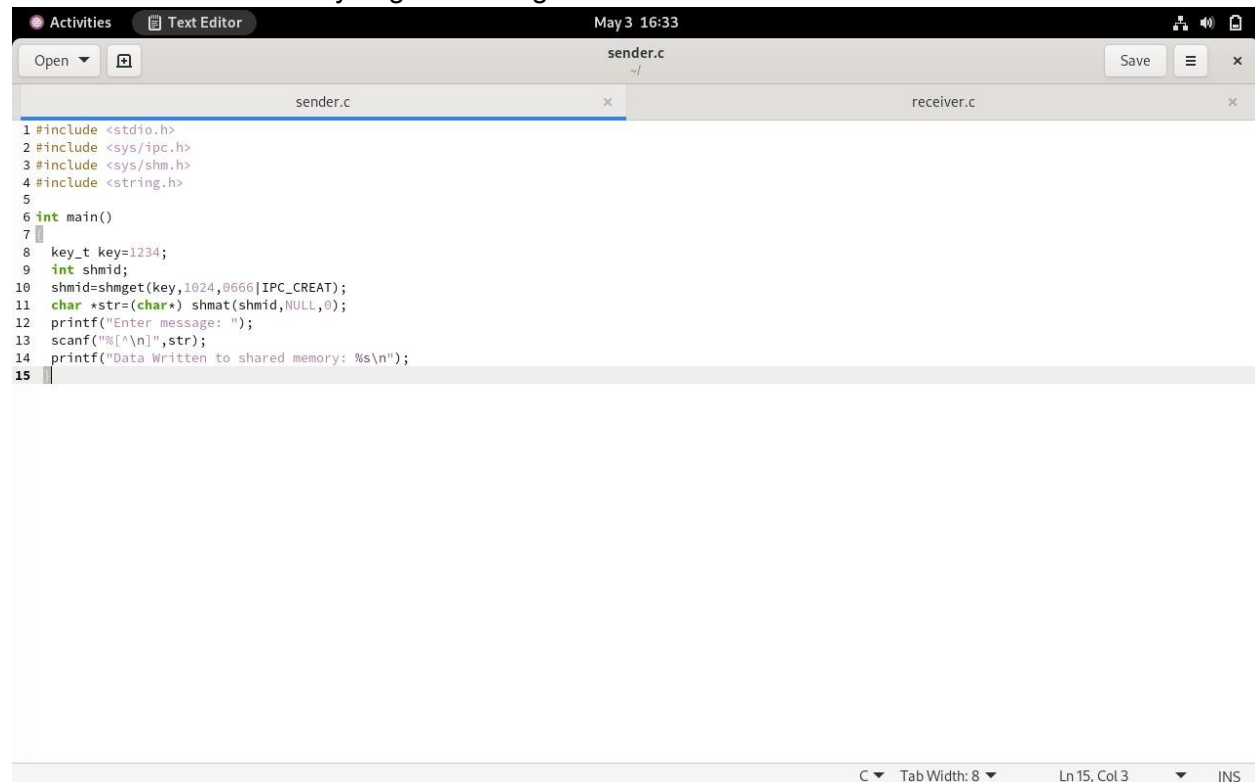
Algorithm:

sender

1. Set the size of the shared memory segment
2. Allocate the shared memory segment using shmget
3. Attach the shared memory segment using shmat
4. Write a string to the shared memory segment using sprintf
5. Set delay using sleep
6. Detach shared memory segment using shmdt

receiver

1. Set the size of the shared memory segment
2. Allocate the shared memory segment using shmget
3. Attach the shared memory segment using shmat
4. Print the shared memory contents sent by the sender process.
5. Detach shared memory segment using shmdt



The screenshot shows a text editor window titled "Text Editor" with a timestamp of "May 3 16:33". The editor has two tabs: "sender.c" and "receiver.c". The "sender.c" tab is active, displaying the following C code:

```
1 #include <stdio.h>
2 #include <sys/ipc.h>
3 #include <sys/shm.h>
4 #include <string.h>
5
6 int main()
7 {
8     key_t key=1234;
9     int shmid;
10    shmid=shmget(key,1024,0666|IPC_CREAT);
11    char *str=(char*) shmat(shmid,NULL,0);
12    printf("Enter message: ");
13    scanf("%[^\n]",str);
14    printf("Data Written to shared memory: %s\n");
15 }
```

The status bar at the bottom indicates "C", "Tab Width: 8", "Ln 15, Col 3", and "INS".

ActivitiesText EditorMay 3 16:34

receiver.c

sender.creceiver.c

Name: ~/sender.c
MIME Type: plain text document (text/plain)
Encoding: Unicode (UTF-8)

```
1  #include <stdio.h>
2  #include <sys/types.h>
3  #include <unistd.h>
4  #include <sys/shm.h>
5
6
7  key_t key=1234;
8  int shmid;
9  shmid=shmget(key,1024,0666);
10 char *str=(char*) shmat(shmid,NULL,0);
11 printf("Data read from shared memory: %s\n",str);
12 shmdt(str);
13 }
```

C Tab Width: 8 Ln 7, Col 18 INS

```
[PraveenKumar212@localhost ~]$ gcc sender.c -o sender
[PraveenKumar212@localhost ~]$ ./sender
Enter message: Hello OS
Segmentation fault (core dumped)
[PraveenKumar212@localhost ~]$ gcc receiver.c -o receiver
[PraveenKumar212@localhost ~]$ ./receiver
Data read from shared memory: Hello OS
[PraveenKumar212@localhost ~]$
```